

1 Latent-class and mixture-model formulations for
2 biomarker-treatment interaction in N-of-1 trials

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5 **Contents**

6	1 Abstract	2
7	2 Introduction	3
8	2.1 The aggregated N-of-1 trial and the biomarker-moderation question	3
9	2.2 Two formal answers: continuous heterogeneity and latent-class structure	3
10	2.3 Related work: two methodological traditions	4
11	2.4 Why this matters for N-of-1 designs specifically	5
12	2.5 Aims and structure of the paper	6
13	3 Notation	6
14	4 Faithful latent-class data-generating processes	7
15	5 MVN approximation to mixture-induced moderation	7
16	6 Failure regimes of the MVN approximation	8
17	7 Implications for power under carryover	8
18	8 Identifiability at trial-relevant sample sizes	9
19	9 Finite-mixture taxonomy	9
20	10 The biomarker-moderation spectrum	10
21	11 Implications for analysis and design	11
22	12 Software for mixture analyses	12
23	13 Discussion	12
24	13.1 Progress to date	13
25	13.2 Pilot results (Study 5, 240 replicates)	14
26	14 Conflict of interest	14
27	15 Funding	15

28	16 Data availability statement	15
29	17 Reproducibility	15
30	18 References	15

31 1 Abstract

32 **Background.** Aggregated N-of-1 clinical trials test biomarker-treatment interactions through
33 a fixed continuous- heterogeneity regression in published practice. The psychometric litera-
34 ture has developed mature finite-mixture and latent-class machinery for representing discrete
35 subpopulation structure that the biostatistical literature on N-of-1 trials has not engaged.
36 The choice between continuous-heterogeneity and latent-class encodings of the biomarker-
37 moderation question is a substantive scientific question rather than a parameterization con-
38 venience, but the field has not characterized when class-aware analyses become competitive
39 with continuous regression at trial-relevant sample sizes.

40 **Methods.** We develop a five-component biomarker-moderation spectrum (covariance-only,
41 mean-and-covariance, location-scale, heterogeneous random-effects, latent-class) bridging
42 continuous-heterogeneity and latent-class formulations, and we catalogue the finite-mixture
43 taxonomy (latent class analysis, growth mixture models, factor mixture models, regression
44 mixtures) for application to within-subject longitudinal designs. We commit a heterogeneous-
45 random-slopes generative data-generating process (DGP) and a class-aware analysis procedure
46 (`lcmm::hlme` with gridsearch initialization), and run a 240-replicate-per-cell prototype simu-
47 lation at $N = 70$, gating slope $\in \{0, 1.0, 1.5\}$, on a 4-core parallel implementation. Three test
48 forms are recorded per replicate: the linear-mixed `bm:Dbc` Wald, the unconditional Wald on
49 the `lcmm` gating coefficient, and a BIC-conditional gating procedure.

50 **Results.** In a 240-replicate-per-cell pilot we find that the unconditional Wald test on the
51 latent-class gating coefficient inflates Type I error to 0.197 (95% CI [0.145, 0.250]) at the
52 null, decisively above nominal 0.05 and consistent with the¹ and² overextraction predictions.
53 The BIC-conditional gating procedure controls Type I (0.004 under the null) but is severely
54 under-powered at trial-relevant N (rejection rate 0.025 at gating slope 1.0, 0.004 at 1.5). The
55 linear-mixed `bm:Dbc` Wald test holds nominal Type I and reaches 0.542 power at gating slope
56 1.5, dominating both class-aware tests. The Monte Carlo standard error (MCSE) on power is
57 approximately 0.030 across the grid. These pilot estimates precede the production five-study
58 program described in the companion simulation plan, which has not yet been executed.

59 **Conclusions.** Within the 240-replicate pilot, at trial- relevant sample sizes ($N = 70$, eight
60 measurement timepoints), neither the unconditional Wald-on-gating nor the BIC-conditional
61 gating test is competitive with the linear-mixed baseline under the heterogeneous-random-
62 slopes DGP. The latent-class encoding appears to require a regime of large class separation
63 for class-aware tests to gain power; the pre-registered production program (five studies, $N \in$
64 $\{35, 70, 100\}$, wider gating- slope grid) has not yet been executed and is required to map the
65 regime where the BIC-conditional procedure becomes competitive.

2 Introduction

2.1 The aggregated N-of-1 trial and the biomarker-moderation question

The aggregated N-of-1 trial design is a class of clinical trials in which each enrolled participant cycles between active treatment and control across multiple within-person blocks, with within-person comparisons pooled across participants for population-level inference^{3–6}. The design preserves the within-subject contrast that gives single-participant N-of-1 trials their inferential strength while accumulating the statistical efficiency of a multi-participant trial. For chronic conditions whose treatment response is reversible on a useful timescale, the design yields more information per participant per unit time than a parallel-group randomized trial of equivalent total sample size⁷. It has been deployed in chronic pain⁸, inflammatory bowel disease, neuropsychiatric symptoms in post-traumatic stress disorder⁹, and a growing portfolio of rare and orphan disease applications where parallel-group trials are infeasible^{5,10}.

Within this design class, a question of increasing scientific weight is whether treatment response is moderated by a measured pre-treatment biomarker, allowing clinicians to stratify treatment decisions at the point of care under the formal framework laid out in the PATH statement¹¹. The corresponding statistical estimand is the biomarker-treatment interaction term in the within-participant outcome model. Power to detect this interaction depends on the magnitude of moderation, the precision of the within-participant contrasts, the trial design (open-label, crossover, hybrid), and any time-dependent leakage between treatment phases, conventionally termed ‘carryover’, that erodes the cleanness of the within-subject contrasts.

Recent methodological work has documented substantial power losses for biomarker-treatment interaction inference under realistic carryover, with the loss varying sharply by how the moderation effect is represented in the data-generating process⁹. The present paper takes up a question that the prior work raised but did not resolve: how should biomarker moderation be represented in the DGP itself? Two formal answers, drawn from a long history of work outside biostatistics, dominate the available options. We begin by laying out these two answers and the methodological tradition behind each, before turning to the specific consequences for N-of-1 trial design and analysis.

2.2 Two formal answers: continuous heterogeneity and latent-class structure

The first answer represents biomarker moderation as a graded continuous effect: every participant has some degree of treatment response, the response is a smooth (typically linear) function of the biomarker, and population heterogeneity is captured by a single multivariate distribution with an interaction parameter. This is the dominant tradition in clinical biostatistics, where heterogeneity of treatment effect is routinely modeled via random slopes on the treatment indicator and via interaction terms in fixed-effects regression. In its most economical form, the moderation enters the conditional mean of the outcome as $\beta_{bm:D} \cdot B \cdot D$, where B is the biomarker and D is the (possibly continuous-decay) treatment indicator.

The second answer represents biomarker moderation as a discrete subpopulation structure:

participants belong to one of a small number of unobserved classes (most commonly ‘responder’ and ‘non-responder’), the classes have qualitatively distinct treatment-response curves, and the biomarker predicts class membership without determining it. The corresponding generative form is a finite mixture in which class membership is a Bernoulli or multinomial draw with mixing probabilities that depend on the biomarker, and the within-class outcome model is class-specific. The marginal moderation observed in the data emerges from the mixture rather than from a smooth conditional-mean function.

These two answers correspond to genuinely different biological hypotheses. The continuous-heterogeneity view is consistent with a mechanism in which the biomarker reflects a graded physiological substrate (receptor density, enzymatic activity, baseline trait severity) that scales drug effect across the entire population. The latent-class view is consistent with a mechanism in which qualitatively distinct phenotypes exist (a metabolising versus non-metabolising allelic variant; a present versus absent target receptor) and the biomarker is an imperfect indicator of phenotype. The two views can produce identical marginal biomarker-outcome associations at a single timepoint and diverge sharply once the longitudinal structure of an N-of-1 trial is engaged, because carryover and within-person serial dependence interact differently with continuous and class-structured heterogeneity. Both traditions have long histories of formalization and identifiability analysis in their respective fields, and the corresponding tools are mature.

2.3 Related work: two methodological traditions

The continuous-heterogeneity and latent-class answers each arrive with a distinct history that this section summarizes; the full literature review, tracing both traditions decade by decade, is given in the companion supplement (`supplement.pdf`, Section 1).

The continuous-trait view originates with Spearman’s two-factor theory of intelligence¹² and was elaborated through the modern factor-analytic tradition^{13,14}, in which between-person heterogeneity in a latent construct is by assumption continuous. The discrete-class view emerged from Lazarsfeld’s latent structure analysis^{15,16}, placed on rigorous footing by Goodman¹⁷ and unified in the finite-mixture monograph of McLachlan and Peel¹⁸. Bartholomew¹⁹ framed the tension between the two views as a choice between ‘graded’ and ‘unfolding’ models that the data themselves rarely settle without strong prior assumptions.

A largely independent strand developed in econometrics and marketing. Heckman and Singer²⁰ showed that structural parameter estimates in duration models depend sensitively on the assumed heterogeneity distribution and proposed estimating that distribution as a discrete mixture, and Lindsay²¹ placed the resulting theory of mixture likelihoods on a unified footing. In marketing science, DeSarbo and Cron²², Wedel and DeSarbo²³, and Jedidi and colleagues²⁴ developed covariate-dependent gating functions for segment membership, a structure formalized independently in machine learning as ‘mixtures of experts’²⁵. This gating-function machinery is the exact structure the biomarker-moderation problem requires: whether a single observed covariate (the biomarker) predicts latent class membership, and whether the within-class treatment coefficient differs across classes.

145 By the late 1990s both strands converged on hybrid factor mixture models^{2,26,27} and their
146 longitudinal extension, the growth mixture model^{28–32}. Bauer and Curran¹, Bauer³³, Lubke
147 and Muthen², and Nylund and colleagues³⁴ subsequently documented overextraction hazards:
148 growth and factor mixture models fitted to single-component, non-gaussian data can spuri-
149 ously identify multiple latent classes, and reliably distinguishing genuine from spurious class
150 structure typically requires several hundred to several thousand participants. This caution is
151 central to the identifiability argument developed below.

152 Mixture methods entered longitudinal biostatistics through Verbeke and Lesaffre³⁵ and Zhang
153 and Davidian³⁶, and through the `lcmm`³⁷ and `flexmix`^{38,39} R packages used in the present
154 paper’s pilot. Despite this translation, mixture and latent-class methods have not, to the best
155 of our knowledge, been systematically applied to aggregated N-of-1 trial designs: the published
156 N-of-1 methods literature specifies biomarker moderation as a continuous interaction term
157 throughout^{5,6,9,10,40–42}, with no parallel treatment of the latent-class formulation. This gap is
158 what the present paper addresses.

159 2.4 Why this matters for N-of-1 designs specifically

160 Three features of the aggregated N-of-1 design make the continuous-versus-discrete modera-
161 tion question particularly consequential for design and analysis decisions. First, the within-
162 participant longitudinal structure of an N-of-1 trial provides repeated measurements at multi-
163 ple drug states (on-drug, off-drug, in carryover). Under a continuous-heterogeneity DGP the
164 moderation signal is carried by how the within-participant treatment contrast varies with the
165 biomarker; under a latent-class DGP the signal is carried by class membership, which is fixed
166 across all measurements within a participant and is therefore in principle recoverable from the
167 full trajectory even when individual on-drug contrasts are noisy. The two formulations imply
168 different efficient analysis strategies and different power profiles under carryover⁹. Second,
169 N-of-1 trials are typically conducted at $N \in [30, 150]$, well below the sample sizes at which
170 mixture-model identifiability has been validated in the psychometric literature, so the iden-
171 tifiability cautions documented by¹ and² apply with full force: even if a latent-class DGP is
172 the biologically correct generative model, mixture analysis at trial-relevant sample sizes may
173 be underpowered or unidentified, and a misspecified single-component analysis may be the
174 more reliable inferential tool despite its bias. Resolving this trade-off is in part an empirical
175 question the existing psychometric simulation literature has not addressed at N-of-1-relevant
176 sample sizes. Third, carryover, which enters the analysis as a continuous decay in the effective
177 drug indicator $D_{bc,it} = \exp(-\lambda t_{sd,it})$, interacts differently with the two encodings: it erodes
178 the moderation signal directly in the continuous-heterogeneity formulation, where modera-
179 tion enters as a product with $D_{bc,it}$, but erodes only the within-class drug contrast in the
180 latent-class formulation, leaving class-membership information available if it can be recovered
181 from the full trajectory. The two formulations therefore predict different power-loss profiles
182 under carryover, a difference already documented to be empirically large in the companion
183 carryover-sensitivity work.

2.5 Aims and structure of the paper

This paper has four aims: to formalize the latent-class and finite-mixture formulations of biomarker-treatment interaction in aggregated N-of-1 trials, with identifiability conditions specific to the trial-relevant sample-size regime; to characterize estimation strategies and the diagnostics required to distinguish genuine from spurious class structure at small N ; to report a factorial simulation study benchmarking mixture-DGP and continuous-DGP power against the linear-mixed-model and class-aware analyses under realistic carryover; and to apply the developed methods to the prazosin-PTSD biomarker dataset of⁹ in future work.

The remainder of the paper proceeds as follows. Notation fixes terminology; the latent-class generative form, its MVN second-moment approximation, and the regimes in which the approximation breaks down follow in three sections, then the implications for power under carryover and the identifiability constraints that bound mixture analysis at trial-relevant sample sizes. A finite-mixture taxonomy and a biomarker-moderation spectrum locate the relevant model families and DGP encodings in condensed form, with full development in the companion supplement. Implications for analysis and design, and software for mixture analyses, are then drawn together. The discussion addresses three open methodological questions and reports a 240-replicate pilot from the five-study simulation program described in the companion simulation plan, the empirical core of the present line of work.

3 Notation

Throughout, B_i denotes the pre-treatment biomarker value for participant i . BR_{it} denotes the biological-response component of outcome for participant i at time t , decomposed in the Hendrickson framework into three latent response factors: BR_{it} itself, a time-variant component TV_{it} , and a placebo-belief component PB_{it} . $D_{bc,it}$ denotes drug state at time t in the analysis-side continuous-decay form $D_{bc,it} = \exp(-\lambda t_{sd,it})$ during off-drug periods and $D_{bc,it} = 1$ during on-drug periods, with $t_{sd,it}$ the time since drug discontinuation and λ the decay rate. Following the convention used throughout this series, D_{it} is reserved for the binary on-drug state and $D_{bc,it}$ for the continuous exposure-decayed indicator used here; the latter is the Dbc regressor of the analysis model. Z_i denotes an unobserved class label. The half-life of carryover is $t_{1/2} = \ln 2/\lambda$ and is allowed to differ between the DGP and the analysis model.

The principal psychometric and econometric terms used below are:

- **Latent class.** An unobserved categorical class, equivalent in trial vocabulary to an unobserved responder/non-responder class.
- **Latent factor.** An unobserved continuous trait governing covariation among observed variables, equivalent to a frailty (the survival-analysis term for an unobserved multiplicative random effect) or to a random intercept in a linear mixed model.
- **Class-conditional distribution f_z .** The outcome distribution within a single latent class.
- **Mixing proportion π .** The prevalence of a given latent class in the population.

- 223 • **Measurement invariance.** Whether a model applies identically across subpopula-
224 tions.
- 225 • **Growth mixture model (GMM).** A longitudinal mixed model in which the popula-
226 tion of random effects is itself a finite mixture.
- 227 • **Factor mixture model (FMM).** A hybrid model with discrete classes at the upper
228 level and within-class continuous heterogeneity at the lower level.
- 229 • **Regression mixture / mixture of experts.** A finite mixture in which the gating
230 function (class-membership probability) depends on covariates and within-class regres-
231 sions are class-specific.
- 232 • **EM algorithm.** Expectation-maximization (EM) applied to recover unobserved class
233 labels jointly with class-conditional model parameters.

234 4 Faithful latent-class data-generating processes

235 If a candidate predictive biomarker indexes an unobserved responder/non-responder
236 dichotomy, the mechanistically faithful data-generating process is a finite mixture:

$$237 \quad Z_i \mid B_i \sim \text{Bernoulli}(\pi(B_i)), \quad BR_{it} \mid Z_i = z \sim f_z(t, D_{bc,it}),$$

238 where Z_i is the latent class indicator, $\pi(B_i) = \Pr(Z_i = 1 \mid B_i)$ is the class-membership
239 probability (typically a logistic function of the biomarker), and the class-conditional response
240 distributions f_0 and f_1 differ in their drug-response parameters: peak response, onset rate,
241 or both. Within-class temporal structure follows the same modGompertz form used in single-
242 class formulations, with class-specific parameters substituted.

243 Two consequences of this DGP form are central to what follows. First, the class label Z_i
244 is fixed across all measurements within a participant, invariant to drug state and to time.
245 Second, the class-conditional mean trajectory does depend on $D_{bc,it}$: under a full carryover
246 off-drug period ($D_{bc,it} \rightarrow 0$) the two class-conditional means converge if the responder/non-
247 responder distinction is purely in drug response. Class-label invariance therefore preserves
248 participant-level information about the moderation signal that an analysis with access to
249 Z_i could exploit, even where instantaneous between-class mean separation in BR is fully
250 attenuated. The MVN-covariance encoding, in contrast, has no such latent-label information
251 to exploit: the covariance is the entire signal, and its erosion under carryover is direct.

252 5 MVN approximation to mixture-induced moderation

253 A two-component mixture with class-dependent means and common within-class covariance
254 produces a marginal joint distribution of (B_i, BR_{it}) whose first two moments are reproduced,
255 to leading order, by a single multivariate normal with $\text{Cor}(B, BR) = c_{bm}$. Under within-class
256 independence between B_i and BR_{it} ,

$$c_{bm}^2 = \frac{\pi(1-\pi)(\mu_B^1 - \mu_B^0)^2}{\text{Var}(B)} \cdot \frac{\pi(1-\pi)(\mu_{BR}^1 - \mu_{BR}^0)^2}{\text{Var}(BR)},$$

so that c_{bm}^2 is the product of the between-class variance fractions in B and in BR . The marginal correlation that an MVN can reproduce is therefore bounded by the geometric mean of the two intra-class correlations; if either class separation is small, the achievable marginal c_{bm} is small even when class structure is otherwise well-defined.

Under mild regularity conditions (adequate class overlap and mixing proportions π bounded away from 0 and 1), the mixture's second-moment structure is well-approximated by a single MVN with differential correlation across drug states: elevated correlation where class-conditional means separate (on-drug) and shrinkage towards zero where they coincide (off-drug after carryover has decayed). For power calculations driven by t-tests on $\beta_{bm:D}$ in a linear mixed-effects model, this approximation is adequate within the regularity conditions and captures the covariance consequences of latent-class structure without committing to a discrete generative form.

6 Failure regimes of the MVN approximation

The MVN approximation is misleading in three identifiable regimes, each with a distinct empirical signature. Under **bimodal responses**, when f_0 and f_1 are well-separated, the marginal distribution of BR is genuinely bimodal, no MVN matches its tails, and power under the true mixture DGP will exceed the MVN-LMM prediction because a well-specified mixture analysis can exploit the bimodality; the pattern occurs where a subpopulation genuinely does not respond, as in certain targeted oncology indications. Under a **strong class-membership gradient**, when $\pi(B_i)$ is steep and approaches a step function at a biomarker threshold, the biomarker behaves nearly deterministically as a class label and a thresholded mean-moderation specification becomes the better approximation, corresponding clinically to a predictive biomarker with near-perfect sensitivity and specificity. Under **class-varying covariance**, if autocorrelation, residual variance, or placebo-response parameters differ between classes, no single-component MVN can represent the generative structure and analysis models assuming constant covariance are misspecified regardless of whether the biomarker interaction is represented in the mean or the covariance; this is the latent-class analogue of heteroscedastic residual variance addressed in standard mixed-effects practice with `varIdent` or `weights` specifications, except that the stratifying variable is unobserved.

7 Implications for power under carryover

The substantial covariance-only power loss under carryover (40–60% across designs in⁹) reflects erosion of a second-moment signal. A mixture-model analysis fitted to mixture-DGP data can in principle recover additional power by estimating class membership directly, because the class label is carryover-invariant. The gap between the covariance-only carryover-sensitive power estimate and the mixture-faithful power estimate upper-bounds what a mixture anal-

293 ysis could recover; the bound is asymptotic, and at finite N finite-sample bias can drive the
 294 realized gap below the bound or, in adverse identifiability regimes, reverse its sign. A fo-
 295 cused factorial simulation isolating second-moment approximation error from analysis-model
 296 mis-specification is described in the companion simulation plan (Study 1: mixture-vs-MVN
 297 factorial), crossing mixture and MVN DGPs with linear-mixed and joint latent-class-mixed
 298 analyses; the mixture-DGP/MVN-analysis versus mixture-DGP/mixture-analysis contrast
 299 quantifies power recoverable by a faithful analysis model, and a null check on the MVN-
 300 DGP/mixture-analysis cell verifies the mixture model does not extract spurious power where
 301 no latent-class structure exists.

302 8 Identifiability at trial-relevant sample sizes

303 Mixture modeling is attractive on faithfulness grounds but carries identifiability costs that
 304 are acute in N-of-1 settings. The EM algorithm for finite mixtures of mixed-effects mod-
 305 els requires either informative priors on class-membership probabilities or strong separation
 306 between class-conditional response curves, and the standard diagnostic for mixture identifi-
 307 ability, the entropy of posterior class-membership probabilities, requires substantial sample
 308 sizes (N several hundred to several thousand) to stabilize at values supporting confident class
 309 extraction in published psychometric applications^{1,33,34}. Aggregated N-of-1 trials of the size
 310 contemplated in current biomarker applications, $N \in [30, 150]$, are well below this range, so
 311 mixture analysis at trial-relevant sample sizes may be underpowered or unidentified even when
 312 the underlying DGP is genuinely mixture-structured. The question is therefore not whether
 313 mixture formulations are more faithful, but whether the fidelity gain is recoverable at realistic
 314 sample sizes, or whether analysis-strategy mitigations within standard mixed-effects practice
 315 (restricted analysis, weighting, within-subject contrasts; see⁹), which have better-understood
 316 inferential properties at the relevant sample sizes, deliver a better power-per-complexity trade.
 317 Resolving this empirically requires the simulation program described in the companion plan;
 318 Study 5 addresses small- N identifiability and Type I error directly.

319 9 Finite-mixture taxonomy

320 Four families of finite mixture model from the psychometric and econometric literatures bear
 321 directly on the biomarker-treatment interaction problem. Table 1 summarizes each family's
 322 structure, origin, and relevance to the N-of-1 setting; the full development of each, including
 323 the reasoning behind the package recommendations, is given in the companion supplement
 324 (Section 2).

Table 1: Finite-mixture model families relevant to the biomarker-moderation problem.

Family	Structure	Origin	Relevance to N-of-1 trials
Latent class / latent profile analysis (LCA/LPA)	Cross-sectional mixture; class-specific marginal distributions	^{16,17,18}	Baseline responder-class discovery from biomarker panels; poLCA
Growth mixture model (GMM)	Longitudinal mixed model with a class-mixture random-effects population; class label is drug-state-invariant	^{28,31}	Closest analogue of the faithful latent-class DGP in Section 3; overextraction cautions ¹ apply with full force at N-of-1 sample sizes
Factor mixture model (FMM)	Discrete classes at the upper level, continuous within-class factor structure at the lower level	^{26,27,2}	Biomarker as both a noisy class indicator and a continuous within-class modulator; higher estimation cost
Regression mixture / mixture of experts	Class-specific regression surfaces with a covariate-dependent gating function	^{22,25}	Biomarker governs both class probability and within-class drug-effect magnitude; flexmix

GMM is the closest psychometric analogue of the faithful latent-class generative form developed in Section 3: each participant belongs to a latent class with its own drug-response trajectory, and a GMM that uses the biomarker as a class-membership predictor is essentially that generative model written in psychometric vocabulary. This is why the overextraction cautions from the GMM literature^{1,33} carry directly into the identifiability discussion below.

10 The biomarker-moderation spectrum

The covariance-only formulation of⁹ specifies a biomarker-dependent covariance structure with population means held constant across biomarker values. Locating this formulation within the broader spectrum of biomarker-moderated DGPs is useful because the moment-by-moment organization of the spectrum (biomarker moderating the mean, the covariance, both, or higher moments) clarifies which features of the carryover-induced power loss are properties of the

specification itself rather than of the underlying scientific question.

Table 2 organizes the five points on the spectrum moment by moment; the full development of each, including the measurement-invariance parallel for the covariance-only case and the GAMLSS distributional-regression formalism, is given in the companion supplement (Section 3).

Table 2: The biomarker-moderation spectrum, organized by which moment of the response distribution carries the biomarker signal.

Point on spectrum	What varies by class/biomarker	Behaviour under carryover
Covariance-only ⁹	$\text{Cov}(B, BR)$ only; means constant	Signal is entirely in the eroded channel; power loss is largest
Combined mean and covariance ²⁷	Both $E[BR B]$ and $\text{Var}[BR B]$	Mean channel survives carryover; power loss intermediate
Location-scale (GAMLSS) ^{43,44}	Location and scale of the marginal distribution, no discrete classes	Analogue of the combined case without committing to a class count
Heterogeneous random-effects ^{35,37}	Class-specific random-intercept/slope distributions	Class label is carryover-invariant; used for the Study 5 pilot below
Pure mean moderation (Architecture A)	$E[BR B]$ only; covariance constant	Signal survives carryover essentially undiminished

The covariance-only specification of⁹ is the most restrictive point on this spectrum: it holds $E[BR_{it}]$ fixed across biomarker strata and places the entire interaction in $\text{Cov}(B_i, BR_{it} | D_{bc,it})$, which is an unusual restriction within the psychometric mixture tradition (most standard FMM and GMM parameterizations permit class-specific means at a minimum). Its appeal is tractability for power simulation under multivariate normality; its cost is that it does not correspond cleanly to a natural generative mechanism, and, because the entire signal sits in a channel that carryover erodes by construction, a sensitivity analysis based solely on this point risks overstating the carryover-induced power loss a biomarker-guided trial should actually expect. The heterogeneous random-effects row is the point used for the Study 5 pilot reported below, because it preserves compatibility with the existing `nlme::lme` analysis machinery while admitting a biologically faithful latent-class generative structure at the DGP level; the `lcmm` package³⁷ implements this family directly.

11 Implications for analysis and design

Three implications follow from locating the covariance-only encoding within the broader spectrum. First, the covariance-only parameterization is the most restrictive operational case of

a much richer family: the substantial power loss under carryover documented in⁹ is a specific property of this restrictive case, because it places the entire biomarker-outcome signal in a channel carryover erodes by construction, and need not generalize to the mean-plus-covariance or heterogeneous-random-effects variants standard in psychometric and biostatistical practice. A complete sensitivity analysis should report power under at least three points on the spectrum: pure mean moderation, pure covariance moderation, and a combined specification. Second, the identifiability concerns documented in the GMM and FMM literatures^{1,2,33,34} bound what can be asked of mixture-based analysis models at N-of-1 sample sizes: published FMM applications typically involve several hundred to several thousand participants, an order of magnitude above biomarker-trial applications with $N \in [30, 150]$, so analysis plans built around fitting `lcmm` or `flexmix` directly to trial data should not be relied on as the primary analysis, though mixture analysis retains value as a sensitivity check and as a DGP for power simulation. Third, if N-of-1 simulation infrastructure pursues a richer DGP in subsequent work, the natural target is the heterogeneous random-slopes form rather than a full FMM, because it preserves the analysis model's compatibility with the existing `nlme::lme` machinery while admitting a biologically faithful latent-class generative structure at the DGP level, preserving the audit chain of the existing framework and permitting direct comparison against the covariance-only baseline under matched effect-size tuning.

12 Software for mixture analyses

R packages relevant to estimating the models surveyed above, in approximate order of fit to the longitudinal aggregated N-of-1 use case: `lcmm`³⁷, the closest available tooling for heterogeneous random-effects specification, with syntax closely related to `nlme::lme`; `flexmix`^{38,39}, a more general regression-mixture framework requiring more configuration; `mclust`⁴⁵, for cross-sectional latent-profile analyses of baseline biomarker panels; `gamlss`⁴³, for distributional regression; `poLCA`⁴⁶, for categorical-indicator LCA; and `OpenMx` and `lavaan` with mixture extensions, for FMM-style structural-equation specifications at higher configuration cost. The reference implementation outside R is Mplus⁴⁷, the standard environment for GMM, FMM, and related mixture-SEM work, and the environment used by much of the identifiability and overextraction literature cited above^{1,2,34}.

13 Discussion

The biomarker-treatment moderation problem in aggregated N-of-1 trials sits at the intersection of two methodological traditions that have, until now, only weakly engaged with each other: a biostatistical tradition with efficient within-participant estimators but a continuous-interaction-only specification of moderation, and a psychometric tradition with mature finite-mixture machinery applied primarily at sample sizes well above the N-of-1 range. The argument of this paper is that the choice between continuous-heterogeneity and latent-class encodings is a substantive scientific question, not a parameterization convenience, opening both a sensitivity-analysis program (the moderation spectrum surveyed above) and an analysis-strategy program (class-aware mixture analyses where identifiability permits, mitiga-

395 tions where it does not).

396 Two limitations bound what the paper currently provides. First, the 240-replicate pilot re-
397 ported below is a single cell of the five-study production factorial (mixture-vs-MVN, the
398 mean-covariance-combined triptych, heterogeneous random slopes, the three-regime break-
399 down grid, and the small- N identifiability pre-flight) described in the companion simulation
400 plan; the production run at 1,000+ replicates per cell has not yet been executed. Second,
401 the application to real biomarker-guided trial data has not been carried out; the prazosin-
402 PTSD analysis of⁹ provides the most immediate target for re-analysis under at least three
403 biomarker-moderation encodings once the production run completes.

404 Three open methodological questions remain, to be addressed by the production program.
405 The first is whether class-aware analyses are usable as primary analyses at any realistic N -
406 of-1 sample size, or only as sensitivity checks; the pilot evidence below suggests the latter,
407 and only the wider production grid can settle it definitively. The second is how analysis
408 half-life should be specified when the DGP is a mixture rather than a single MVN, since
409 the continuous-decay drug indicator $D_{bc,it} = \exp(-\lambda t_{sd,it})$ has a clean interpretation under
410 a single-component generative model but a less obvious one when class-conditional carryover
411 dynamics differ between classes. The third is whether the trial design itself should respond
412 to the moderation-encoding question: if a latent-class generative form is biologically more
413 plausible than a continuous one, designs that maximize the per-participant contrast and
414 minimize carryover may gain power that the standard hybrid design surrenders.

415 Two secondary theoretical points are developed fully in the companion supplement (Section
416 5). First, the analysis model used throughout this program is a four-term linear mixed model
417 rather than a component-matched non-linear model mirroring the DGP's biological-response,
418 placebo-belief, and time-variant decomposition; the supplement gives four reasons this sim-
419 pler analysis is the right choice at trial-relevant N . Second, the latent-class DGP specified
420 for Study 1 is a natural mechanism for generating apparent subadditivity in a downstream
421 single-component analysis, even absent any direct mechanistic interaction; the Study 1 pre-
422 registration includes an axis designed to distinguish this emergent mechanism from genuine
423 mechanistic interaction empirically.

424 13.1 Progress to date

425 The five-study production program is pre-registered under the ADEMP framework of Morris,
426 White, and Crowther⁴⁸ at `analysis/scripts/latent-class-mixture-application/`
427 `00-ademp-pre-registration.md` (approximately 2,300 cells across the program, seeded
428 random-number control, an effect-size tuning sub-study, and a deviation-log specification).
429 Phase 1 has reached the infrastructure-and-pilot milestone: a class-aware analysis wrapper
430 around `lcmm::hlme` with the same input-output contract as the existing `lme_analysis()`,
431 and a heterogeneous-random-slopes DGP wrapper around `generateData()` that adds
432 a Bernoulli class-membership channel with logistic gating in the biomarker. An earlier
433 100-replicate pre-flight pilot under the null DGP compared six candidate class-aware test
434 forms and found that an unconditional Wald test on the `lcmm` gating coefficient inflates
435 Type I error to five times nominal (0.250 against 0.05), while a BIC-conditional gating

procedure (fit `ng=1` and `ng=2`, test the gating coefficient only if BIC prefers `ng=2`) controls it. That pre-flight comparison, reported in full in the companion supplement (Section 6), fixed the BIC-conditional procedure as the primary class-aware moderation test carried into the production-scale pilot below. Extrapolated to the pre-registered cell counts, the class-aware arm of Study 1 alone is approximately 340 hours of serial CPU time (22–44 hours parallelised across 8–16 cores); the full five-study program is feasible on a single workstation over a week of background compute.

13.2 Pilot results (Study 5, 240 replicates)

The Study 5 prototype was rerun at 240 replicates per cell on 4-core `parallel::mclapply` with three gating-slope levels ($gs \in \{0, 1.0, 1.5\}$) and three test forms (lme `bm:Dbc` Wald, `lcm` gating Wald unconditional, BIC-conditional gating). Wall time was 558 s; convergence was 92.9% under the null and 98.3–98.8% at non-null cells. Per-replicate output is at `analysis/data/quick-sim/03-latent-replicates.rds`; Morris ADEMP summary at `03-latent-summary.txt`. MCSE at the working power near 0.5 is approximately 0.030.

The unconditional Wald-on-gating Type I error of 0.197 (95% binomial CI [0.145, 0.250]) is now decisively distinguishable from the nominal 0.05 (more than five MCSE of separation) and is statistically consistent with the 100-replicate pilot’s 0.250 point estimate. Power at $gs = 1.0$ (0.304) and $gs = 1.5$ (0.360) sits below the lme baseline (0.325 and 0.542 respectively), so the unconditional Wald buys nothing in exchange for the Type I inflation. The BIC-conditional procedure does control Type I (0.004 under the null, with upper CI near 0.013), but its power is 0.025 at $gs = 1.0$ and 0.004 at $gs = 1.5$: under the heterogeneous-random-slopes DGP at $N = 70$, the BIC step selects `ng=1` in essentially every replicate and the conditional procedure approximates a never-reject rule. The non-monotonicity ($gs = 1.5$ below $gs = 1.0$) suggests that as the heterogeneous-slopes signal strengthens, a single-class lme captures it well enough that BIC continues to favor `ng=1` and the conditional procedure does not fire.

The lme `bm:Dbc` test holds nominal Type I (0 of 240 under the null) and reaches 0.542 power at $gs = 1.5$, dominating both `lcm` tests at this sample size and DGP regime. We therefore conclude, on the evidence of this pilot, that at trial-relevant N and under the heterogeneous-random-slopes generating mechanism examined here, neither the unconditional Wald-on-gating nor the BIC-conditional gating test is competitive with the linear-mixed baseline: the unconditional test is severely anti-conservative, and the BIC-conditional test is severely under-powered. A canonical run at 1,000 replicates per cell with a wider gating-slope grid (and at $N \in \{35, 70, 100\}$) is required to map the regime in which the BIC-conditional procedure does become competitive, plausibly at large class separations rather than under heterogeneous random slopes.

14 Conflict of interest

The authors declare no conflicts of interest.

473 15 Funding

474 This work received no specific funding.

475 16 Data availability statement

476 The data and code that support the findings of this study are openly available in the
477 pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-
478 compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are
479 at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility
480 section below.

481 17 Reproducibility

482 This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container
483 (image hash in `Dockerfile`; package set in `renv.lock`). Principal packages are `nlme` (`corCAR1`
484 linear-mixed analysis), `parallel` (4-core `mclapply`), `lcmm` (class-aware analysis, `gridsearch`
485 initialization), and `rmarkdown/bookdown` for the manuscript build. Each simulation driver
486 sets `set.seed(20260507)` at the top of the replicate loop, with per-replicate seeds derived by
487 `+ rep_idx` under `parallel::mc.set.seed`.

488 Per-replicate outputs from the 240-replicate Study 5 pilot are checkpointed at `analysis/data/quick-sim/03-la`
489 with Morris ADEMP cell-level summaries at `03-latent-summary.txt` and the effect-size cali-
490 bration table at `03-calibration-table.rds`. Driver scripts are under `analysis/scripts/latent-class-mixt`
491 including the pilot driver `03-study5-pilot.R` (executed) and the production driver
492 `10-study5-production.R` (committed; not yet run). The pre-registered ADEMP plan is at
493 `00-ademp-pre-registration.md` in the same directory. The manuscript source, bibliogra-
494 phy, and CSL file are at `analysis/report/03-latent-class-mixture-application/`.

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